PHD125 Pharmaceutical Organic Chemistry · Introduction to Organic Molecule

After class: check your understanding

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 25 minutes. Questions follow the five subchapters of the lecture and end with a pharmacy scenario.

Answer without notes first, then use the slides to check. The answer key is at the back.

1.1–1.2 · Organic compounds and their features

Q1 CH1-LO1

Wöhler's 1828 experiment made urea from ammonium cyanate. Why did this matter?

A. It showed organic compounds need a vital force

B. It made an organic compound from an inorganic one, with no living thing

C. It was the first ionic compound made

D. It proved urea is inorganic

Q2 CH1-LO1

Classify each compound (Ball 12.E).

Options: organic · inorganic

C_3H_8O _____

$CaCl_2$ _____

CH_3NH_2 _____

$NaHCO_3$ _____

$C_{12}H_{22}O_{11}$ _____

Q3 CH1-LO1

In which pair does the **first** compound have the higher melting point? (Ball 12.E)

A. NaOH, then CH_3OH

B. CH_3Cl, then KCl

C. C_2H_6, then $CoCl_2$

D. CH_4, then LiH

Q4 CH1-LO1

Every carbon atom in a neutral organic molecule forms four bonds.

True / False

Q5 CH1-LO1

Why are hydrocarbons insoluble in water?

A. They are ionic

B. C–H bonds are nonpolar, so the molecules are nonpolar

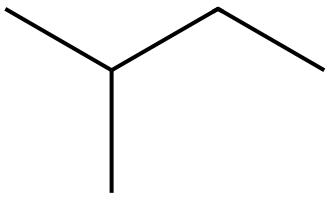
C. They are too heavy

D. They react with water

1.3 · Drawing organic molecules

Q6 CH1-LO2

Write the molecular formula of the compound below. Type it on one line, e.g. C4H10.

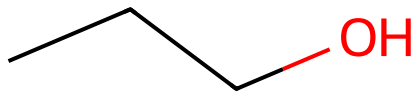


Q7 CH1-LO2

In a line structure, a carbon at a corner with exactly two lines meeting there carries how many hydrogen atoms?

Q8 CH1-LO2

Which is the condensed formula of the compound drawn here?



A. $CH_3CH_2CH_2OH$

B. $CH_3OCH_2CH_3$

C. $CH_3CH(OH)CH_3$

D. CH_3CH_2OH

Q9 CH1-LO2

Match each description to the shape around carbon.

Options: tetrahedral, 109.5° · trigonal planar, 120° · linear, 180°

four single bonds _____

one double bond _____

one triple bond _____

Q10 CH1-LO2

Draw the line structure and the expanded structure of $CH_3CH(CH_3)CH_3$ on paper. Then compare.

1.4 · Functional groups

Q11 CH1-LO3

Match each functional group to its family.

Options: alcohol · carboxylic acid · aldehyde · ether · amide · thiol · ketone

–OH on a chain _____

–COOH _____

–CHO _____

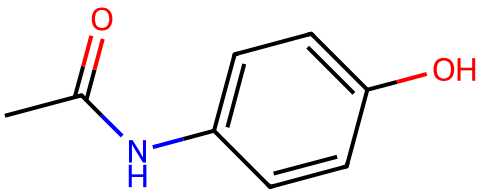
C–O–C _____

$–CONH_2$ _____

–SH _____

Q12 CH1-LO3

Which functional groups are in paracetamol?



A. Aromatic ring, phenol, amide

B. Aromatic ring, alcohol, amine

C. Aromatic ring, carboxylic acid, ester

D. Alkene, phenol, ketone

Q13 CH1-LO3

A C=O group with an –O–C attached to the carbonyl carbon is an ester.

True / False

Q14 CH1-LO3

Benzocaine has an $-NH_2$ on its ring and no C=O next to the N. That group is:

A. an amide
B. an amine
C. a nitrile
D. an alcohol

1.5 · Pharmacy and everyday life

Q15 CH1-LO4

Vitamin C extracted from oranges works better than vitamin C made in a factory because it is natural.

True / False

Q16 CH1-LO4

A customer says a herbal product must be safe because it is natural. The best reply is:

A. Agree: natural products have no side effects
B. Explain that safety depends on the molecules and doses present, not on the source
C. Say natural products never work
D. Say only synthetic drugs contain chemicals

Q17 CH1-LO5

Match each name type for paracetamol.

Options: chemical (IUPAC) name · generic name · brand name

N-(4-hydroxyphenyl)acetamide _____

paracetamol _____

Panadol _____

Q18 CH1-LO5

Why does an old bottle of aspirin smell of vinegar?

A. The aromatic ring breaks open
B. The ester group is hydrolysed by moisture, releasing acetic acid
C. Aspirin turns into ethanol
D. The carboxylic acid evaporates

Q19 CH1-LO5

Morphine and quinine are examples of:

A. synthetic polymers
B. alkaloids from plants
C. inorganic salts
D. esters from animals

Pharmacy scenario · The cold-and-flu shelf

Scenario

Encik Hafiz buys Panadol for a fever. He also picks up a cold-and-flu tablet whose label reads: *paracetamol 500 mg, phenylephrine 5 mg, chlorphenamine 2 mg*. He plans to take both, the full dose of each, for three days.

Q20 CH1-LO5

What is the main problem with his plan?

A. Phenylephrine is an organic compound
B. Both products contain paracetamol, so he could take a double dose
C. Chlorphenamine is a brand name
D. Panadol does not work for fever

Q21 CH1-LO5

If he takes 2 Panadol tablets (500 mg each) four times a day plus 1 flu tablet four times a day, how many grams of paracetamol is that per day?

Q22 CH1-LO5

Write two sentences of advice for Encik Hafiz.

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Adapted from LibreTexts *Introduction to Organic Chemistry – Alkanes* 12.1–12.5 (Sharpe Elles, from Ball, Hill & Scott) and *The Basics of GOB Chemistry* (Ball, Hill & Scott) 12.1, 12.4 and exercises 12.E, CC BY-NC-SA 3.0. Pharmacy items are original material by Dr. NAM. Shared under CC BY-NC-SA for non-commercial teaching.

PHD125 Pharmaceutical Organic Chemistry · Introduction to Organic Molecule

After class: check your understanding

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About 25 minutes. Questions follow the five subchapters of the lecture and end with a pharmacy scenario.

Answer key

1.1–1.2 · Organic compounds and their features

Q1 CH1-LO1

Wöhler's 1828 experiment made urea from ammonium cyanate. Why did this matter?

A. It showed organic compounds need a vital force

B. It made an organic compound from an inorganic one, with no living thing

C. It was the first ionic compound made

D. It proved urea is inorganic

It weakened vitalism, the idea that organic compounds can only come from living things.

Q2 CH1-LO1

Classify each compound (Ball 12.E).

C_3H_8O → organic; $CaCl_2$ → inorganic; CH_3NH_2 → organic; $NaHCO_3$ → inorganic; $C_{12}H_{22}O_{11}$ → organic

Carbon compounds are organic, except CO, CO_2, carbonates, bicarbonates and cyanides. $CaCl_2$ has no carbon.

Q3 CH1-LO1

In which pair does the **first** compound have the higher melting point? (Ball 12.E)

A. NaOH, then CH_3OH

B. CH_3Cl, then KCl

C. C_2H_6, then $CoCl_2$

D. CH_4, then LiH

Ionic compounds (NaOH, KCl, $CoCl_2$, LiH) melt at higher temperatures than the covalent organic compounds.

Q4 CH1-LO1

Every carbon atom in a neutral organic molecule forms four bonds.

True / False → **True**

Carbon is tetravalent: count single, double and triple bonds; they always add up to four.

Q5 CH1-LO1

Why are hydrocarbons insoluble in water?

A. They are ionic

B. C–H bonds are nonpolar, so the molecules are nonpolar

C. They are too heavy

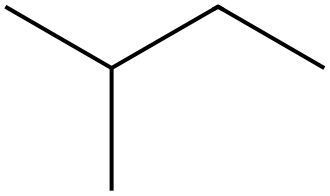
D. They react with water

Carbon and hydrogen have similar electronegativity. Nonpolar molecules do not mix with polar water.

1.3 · Drawing organic molecules

Q6 CH1-LO2

Write the molecular formula of the compound below. Type it on one line, e.g. C4H10.



C_5H_{12}

Five corners and ends = 5 C. Hydrogens: 3 + 1 + 3 + 2 + 3 = 12.

Q7 CH1-LO2

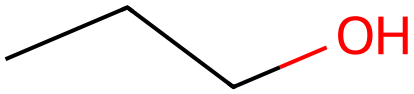
In a line structure, a carbon at a corner with exactly two lines meeting there carries how many hydrogen atoms?

2 H

4 − 2 bonds = 2 H: a CH_2 group.

Q8 CH1-LO2

Which is the condensed formula of the compound drawn here?



A. $CH_3CH_2CH_2OH$

B. $CH_3OCH_2CH_3$

C. $CH_3CH(OH)CH_3$

D. CH_3CH_2OH

Three carbons in a chain with OH on the end carbon: 1-propanol.

Q9 CH1-LO2

Match each description to the shape around carbon.

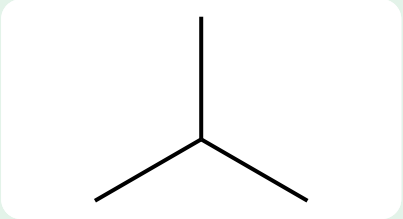
four single bonds → tetrahedral, 109.5°; one double bond → trigonal planar, 120°; one triple bond → linear, 180°

4, 3 or 2 electron groups around carbon.

Q10 CH1-LO2

Draw the line structure and the expanded structure of $CH_3CH(CH_3)CH_3$ on paper. Then compare.

Line structure: a 'Y' shape (three lines from one central carbon). Expanded: a central C bonded to one H and three CH_3 groups, every H drawn. It is 2-methylpropane, C_4H_{10}.



1.4 · Functional groups

Q11 CH1-LO3

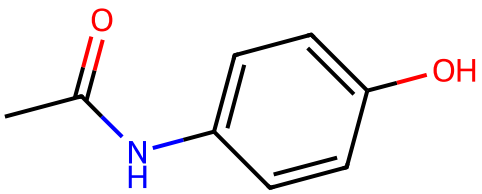
Match each functional group to its family.

–OH on a chain → alcohol; –COOH → carboxylic acid; –CHO → aldehyde; C–O–C → ether; $–CONH_2$ → amide; –SH → thiol

Ketone is the extra option: C=O between two carbons.

Q12 CH1-LO3

Which functional groups are in paracetamol?



A. Aromatic ring, phenol, amide

B. Aromatic ring, alcohol, amine

C. Aromatic ring, carboxylic acid, ester

D. Alkene, phenol, ketone

The –OH sits on the ring (phenol), and the N is bonded to a C=O (amide).

Q13 CH1-LO3

A C=O group with an –O–C attached to the carbonyl carbon is an ester.

True / False → **True**

R–COO–R′ is an ester. With –OH instead it would be a carboxylic acid.

Q14 CH1-LO3

Benzocaine has an $-NH_2$ on its ring and no C=O next to the N. That group is:

A. an amide

B. an amine

C. a nitrile

D. an alcohol

An amide needs N bonded to C=O. Without it, $-NH_2$ is an amine.

1.5 · Pharmacy and everyday life

Q15 CH1-LO4

Vitamin C extracted from oranges works better than vitamin C made in a factory because it is natural.

True / False → **False**

Both are ascorbic acid: the same molecule with the same properties. Only purity, dose and other compounds present can differ.

Q16 CH1-LO4

A customer says a herbal product must be safe because it is natural. The best reply is:

A. Agree: natural products have no side effects

B. Explain that safety depends on the molecules and doses present, not on the source

C. Say natural products never work

D. Say only synthetic drugs contain chemicals

Digoxin from foxglove is natural and toxic in overdose. Everything is made of chemicals.

Q17 CH1-LO5

Match each name type for paracetamol.

N-(4-hydroxyphenyl)acetamide → chemical (IUPAC) name; paracetamol → generic name; Panadol → brand name

The IUPAC name describes the structure; the generic name is used in prescribing; the brand name belongs to a manufacturer.

Q18 CH1-LO5

Why does an old bottle of aspirin smell of vinegar?

A. The aromatic ring breaks open

B. The ester group is hydrolysed by moisture, releasing acetic acid

C. Aspirin turns into ethanol

D. The carboxylic acid evaporates

Aspirin + water → salicylic acid + acetic acid. Acetic acid is the acid in vinegar.

Q19 CH1-LO5

Morphine and quinine are examples of:

A. synthetic polymers

B. alkaloids from plants

C. inorganic salts

D. esters from animals

Both are nitrogen-containing plant compounds: alkaloids.

Pharmacy scenario · The cold-and-flu shelf

Q20 CH1-LO5

What is the main problem with his plan?

A. Phenylephrine is an organic compound

B. Both products contain paracetamol, so he could take a double dose

C. Chlorphenamine is a brand name

D. Panadol does not work for fever

Reading the generic names of all active ingredients shows the duplication.

Q21 CH1-LO5

If he takes 2 Panadol tablets (500 mg each) four times a day plus 1 flu tablet four times a day, how many grams of paracetamol is that per day?

6 g

(2 × 500 + 1 × 500) × 4 = 6000 mg = 6 g, well above the usual adult maximum of 4 g in 24 hours.

Q22 CH1-LO5

Write two sentences of advice for Encik Hafiz.

Take only one paracetamol-containing product at a time, and do not take more than the label dose (no more than 4 g of paracetamol in 24 hours). Always check the generic names of the active ingredients on every medicine label.

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